YASH SONKAR

+91 7008180939 | yash.sonkar@research.iiit.ac.in | Hyderabad, India
linkedin.com/in/yash-sonkar-5a5914227/ | github.com/Yash-Kumar-tech | Google Scholar

TECHNICAL SKILLS

Languages: Python, C++, Rust, Java, Kotlin, SQL, MATLAB
Frameworks & ML Tools: PyTorch, LibTorch, ExecuTorch, Fairseq, TensorRT, Docker
Specializations: Speech-to-Speech (S2S) Synthesis, Streaming Audio Models, Atypical Speech Restoration, Real-time VAD/TTS Pipelines
Tools & Infrastructure: Git, Linux (Ubuntu), Android Studio, LaTeX

ACADEMIC PUBLICATIONS

CLARIS: Clear and Intelligible Speech from Whispered and Dysarthric Voices | *ACM CHI 2026*

- **Authors:** Neil Shah*, **Yash Sonkar***, Shirish Karande, Vineet Gandhi. *(*Equal Contribution)* [Paper Link] | [Project Page]
- Engineered the official inference architecture for an auto-regressive S2U Transformer and Code HiFi-GAN vocoder to convert atypical voices into clear speech.
- Optimized model deployment for production environments by exporting dynamic-axis models to **ONNX** and **OpenVINO** for high-speed CPU execution.

Tongue2Speech: Real-Time Speech Synthesis from Tongue Ultrasound Videos | *Interspeech 2026*

- **Authors: Yash Sonkar***, Yasaswi Kilaru*, Sudheera Yelimeli, Neil Shah, Vineet Gandhi. *(*Equal Contribution)* [Project Page]
- Designed a lightweight framework combining explicit 3D spatio-temporal encoding with a multi-layer Transformer to model long-range articulatory dynamics for Ultrasound-to-Speech.
- Achieved **15.93% WER** (single-speaker) and **31.64% WER** (multi-speaker) on the English TaL corpus, outperforming six strong baseline architectures.

PROJECTS

AI Image Search & Manager | *PyTorch, ChromaDB* | [GitHub]

- Built a desktop application for semantic and hybrid image search leveraging Vision-Language Models (**Qwen3-VL**, **Florence-2**) and **spaCy** NLP tagging.
- Architected a scalable backend combining **ChromaDB** vector embeddings with **SQLite**, utilizing VRAM-aware automated caching to prevent out-of-memory crashes.

Term-Mux: Windows Terminal Multiplexer | *Rust, Windows ConPTY API* | [GitHub]

- Developed an event-driven, asynchronous terminal multiplexer natively for Windows utilizing **Rust** to manage concurrent I/O processes.
- Engineered an internal **ANSI** escape code interpreter and dynamic layout engine to handle real-time side-by-side pane rendering and process termination.

EDUCATION

International Institute of Information Technology (IIIT-H) — Hyderabad, India
Ph.D. in Computer Science and Engineering (CGPA: 8.65/10.00) — *2024 – Present*

- **Research Focus:** Multimodal Speech Generation and Streaming Audio Models
- **Relevant Coursework:** Statistical Methods in AI, Advanced NLP, Computer Vision, Digital Image Processing.

International Institute of Information Technology (IIIT-Bh) — Bhubaneswar, India
B.Tech. in Electronics and Telecommunications Engineering (CGPA: 9.13/10.00) — *2020 – 2024*

- **Relevant Coursework:** Digital Signal Processing, Concepts in Operating Systems, Data Structures in C.